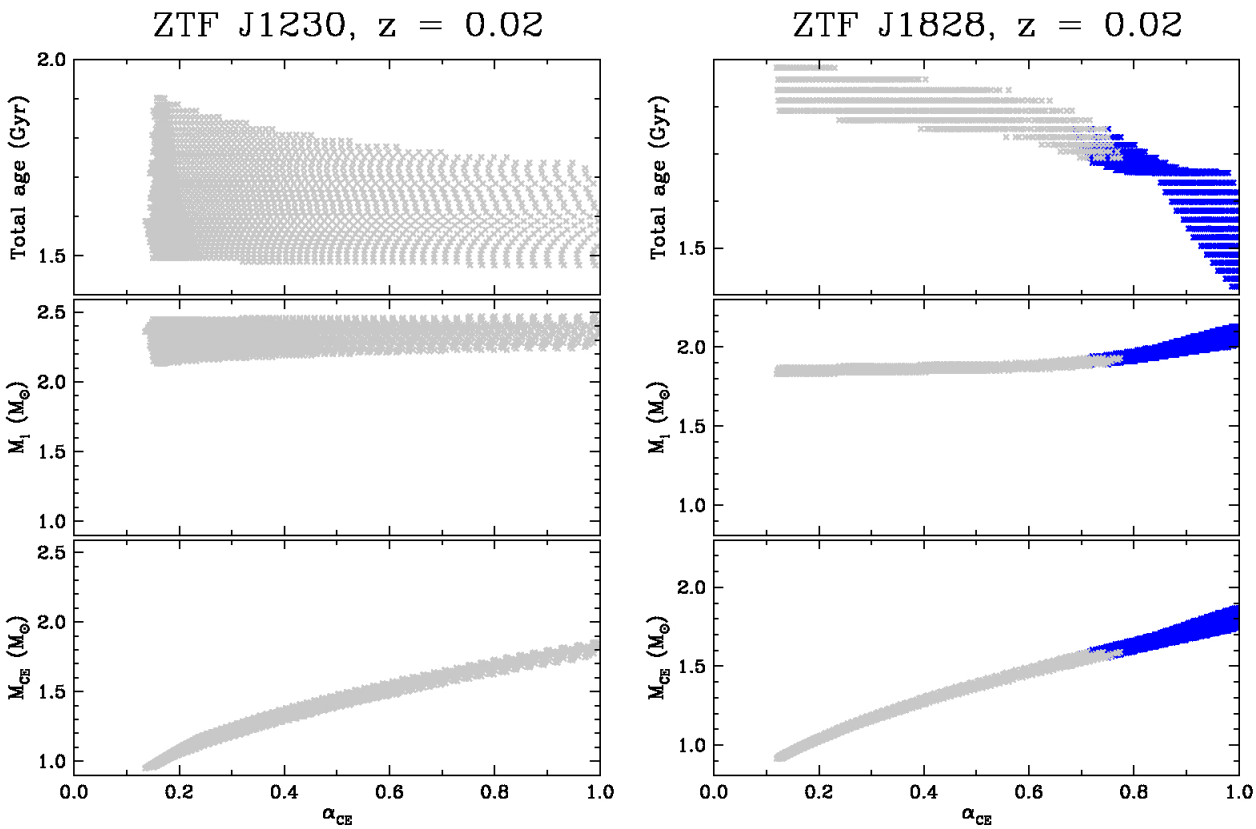

Figure 12. Reconstructions of the evolution of ZTF J1230−2655 (left, assuming a companion mass of 0.021 $M_\odot$) and ZTF J1828+2308 (right), assuming a metallicity of $Z = 0.02$. Shown are the predicted total age of the system (top), the initial mass of the white dwarf progenitor (middle) and the mass of the white dwarf progenitor at the onset of the common envelope phase (bottom), as a function of the common envelope efficiency parameter, $\alpha_{\rm CE}$. Blue points indicate systems where the common envelope occurs during the AGB, before the first thermal pulse, while grey points indicate systems where the common envelope occurs after the first thermal pulse, i.e. during the thermally pulsing (TP)-AGB phase. Only TP-AGB solutions are found for ZTF J1230−2655.

et al. (2024), TP-AGB stars generally transfer mass through winds before they fill their Roche lobe, which tends to increase the mass of the companion before the CE. This process is not included in our reconstruction, but given the low masses of the companions in our binaries, they cannot have accreted a substantial amount of mass. Moreover, it has only been 160 Myr since the CE event in ZTF J1828+2308. This is significantly shorter than the thermal timescale of the brown dwarf, so if it had managed to accrete a large amount of material shortly before the CE phase then we would expect it to still be over-inflated, which it is not. This would seem to argue against a CE phase during the TP-AGB for these systems, making it extremely difficult to understand their evolution. However the Belloni et al. (2024) study was focused on much more massive companion stars, so it is unclear if their conclusions are applicable to these very low mass objects. For example, the mass accreted from winds should strongly depend on the mass and radius of the accretor. Wind mass loss occurs in all directions, so only the fraction of the wind that can be trapped within the Roche lobe of the accretor will be accreted. Even for the solar-type accretors studied by Belloni et al. (2024), the accreted mass is typically only a few percent of a solar mass. Since brown dwarfs are significantly smaller in both mass and radius, it is unlikely that they can trap a considerable amount of wind material. Assuming a similar scaling rate as Bondi-Hoyle-Lyttleton (i.e. the amount of accreted material scales as M^2, Edgar 2004), would imply that these objects accreted around half a Jupiter mass at most during this phase.

To see if it is possible for a CE to occur before the TP-AGB phase in these binaries we repeated our reconstruction but this time assuming very low metallicity ($Z = 0.001$). Low metallicity stars evolve much faster than solar metallicity stars, allowing for less massive progenitors with less envelope mass to expel, hence requiring less orbital energy to be lost from the binary during the CE phase. The results of this are shown in Figure 13. In this case there is a much wider range of possible models which undergo the CE phase before the TP-AGB, including some models with low values of $\alpha_{\rm CE}$. Therefore, if these are both extremely low metallicity systems then their evolutionary history appears to be broadly consistent with other known white dwarf plus brown dwarf binaries. However, the Gaia kinematics of our systems imply that they are both high probability members of the Galactic thin disk, based on the kinematic criteria of Bensby et al. (2003). Stars in this population tend to have much higher metallicities than $Z = 0.001$. Unfortunately, measuring the metallicities of these two systems is not possible at present, but we consider it unlikely that both systems have such low metallicities.

Another possibility for these systems is if there were originally additional objects interior to the brown dwarfs. If these were engulfed shortly before the brown dwarf then they could help facilitate the ejection of the envelope (Bear & Soker 2011; Chamandy et al. 2021), allowing lower mass brown dwarfs to survive a CE phase. However, the small number of known main sequence stars with brown dwarfs in the desert (i.e. within a few AU of the star) do not appear to have

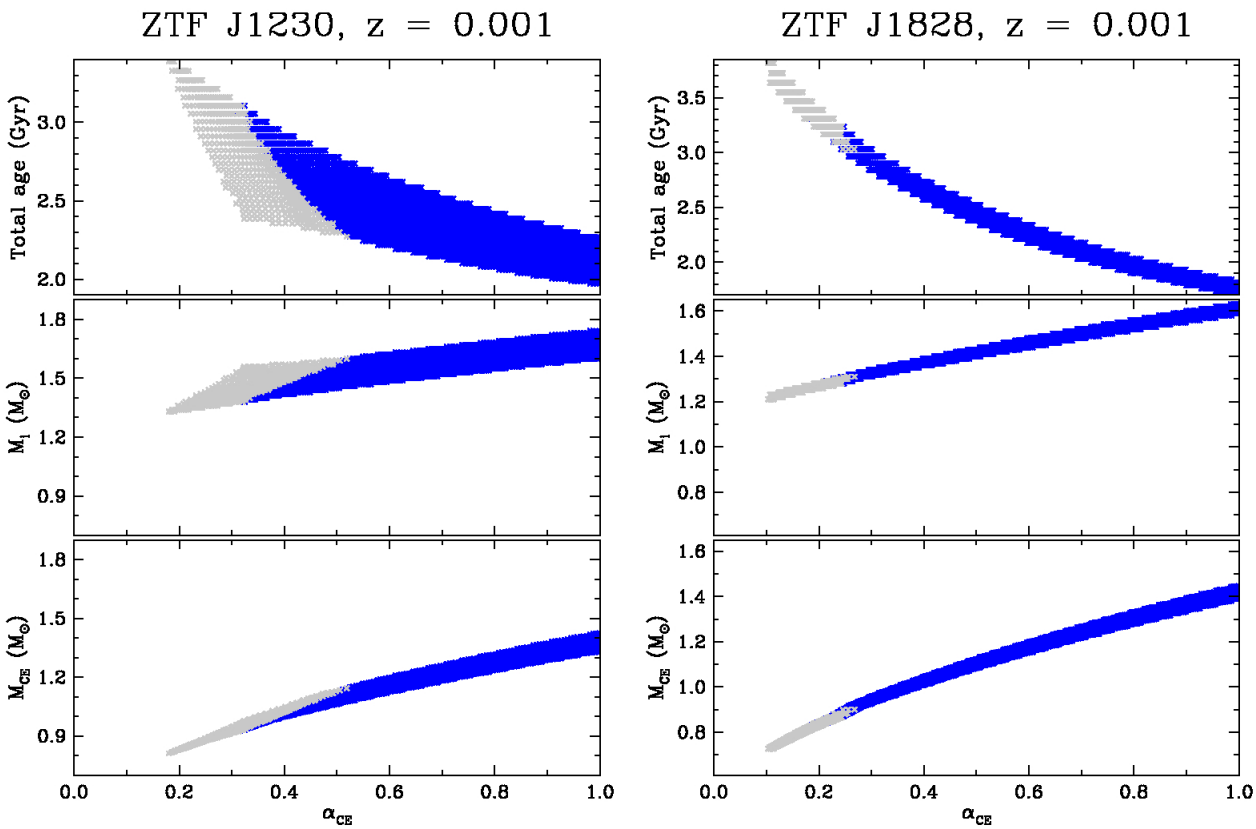

Figure 13. Same as Figure 12, but assuming low metallicity for both systems (Z = 0.001).

a significant number of additional objects in closer orbits. This may be due to the fact that the brown dwarf likely formed further out and migrated into the desert, potentially destabilising any nearby objects as it migrated. Therefore, while this is a potential explanation for these systems (as well as a possible evolution for WD 1856+534; Chamandy et al. 2021), it would require an architecture unlike any system we have currently detected.

Finally, we note the similarities between our findings and the proposed CE evolution for WD 1856+534 (Lagos et al. 2021). Like our systems, the only solutions found for WD 1856+534 required the CE to occur on the TP-AGB, although all solutions required a relatively high value of $\alpha_{CE} > 0.36$. Moreover, Lagos et al. (2021) also required some contribution from additional energy sources to help eject the envelope, which we have not had to include to reconstruct our systems. These differences are partially caused by the lower mass of the object in WD 1856+534, but are primarily the result of the much longer present day orbital period of WD 1856+534 (1.4 days) compared to our systems (0.1–0.2 days). Finding more white dwarf plus brown dwarf binaries with longer periods would be particularly useful, to see if WD 1856+534 is an outlier or consistent with the wider post-CE population.

It may be that these very low mass objects can only survive a CE phase if it occurs right at the end of the evolution of the white dwarf progenitor (i.e. on the TP-AGB), when it has already lost a significant fraction of its original mass and when the envelope is more loosely bound. In this case we would expect the lowest mass brown dwarfs to be only found around CO (or ONe) core white dwarfs. Any similar mass brown dwarf found in a close binary with a He core white dwarf or sdB star would likely pose a significant challenge to reconstruct, since the CE phase would have occurred much earlier, on the FGB. Hydrodynamical simulations place a minimum mass for the companion to eject the envelope at this evolutionary stage of around 30 M_{Jup} (Kramer et al. 2020), well above the brown dwarf masses measured in our systems.

5 CONCLUSIONS

We have identified two very low mass substellar objects in close, eclipsing orbits around white dwarfs. Using a combination of phase resolved X-shooter spectroscopy and high speed multi-band ULTRACAM and HiPERCAM imaging data we have measured the stellar and binary parameters for these two systems, finding that ZTF J1828+2308 contains a 19.5 M_{Jup} brown dwarf in a 2.7 hour orbit around a hot (15900 K) CO core (0.61 $M_\odot$) white dwarf and ZTF J1230−2655 contains a <22.1 M_{Jup} brown dwarf in a 5.7 hour orbit around a cool (10000 K) CO core (0.65 $M_\odot$) white dwarf. We found that the mass and radius of the brown dwarf in ZTF J1828+2308 are consistent with evolutionary models, but the brown dwarf in ZTF J1230−2655 is significantly over-inflated, despite it being the less irradiated of the two. We also reconstructed the past evolution of these binaries, finding that in both cases the common envelope phase likely occurred at the very end of the evolution of the white dwarf progenitor, after the first thermal pulse and when the progenitor had already lost a significant fraction of its original mass. A similar evolutionary history was also proposed for the transiting planet candidate around WD 1856+534, potentially

indicating that common envelope events at this evolutionary stage may allow very low mass brown dwarfs or even planetary mass objects to survive a common envelope phase. Alternatively, both our systems could have very low metallicities, allowing a more standard common envelope phase before the first thermal pulse, or there may have originally been additional bodies interior to the brown dwarf, which were engulfed shortly before the brown dwarf, helping to eject the envelope. Nevertheless, it is clear that post-common envelope binaries with extremely low mass components offer a powerful test of common envelope evolution.

ACKNOWLEDGEMENTS

The results presented in this paper are based on observations collected at the European Southern Observatory under programme IDs 106.D−0824 and 113.D−0277 and on observations made with the Gran Telescopio Canarias (programme ID GTC119-23B), installed in the Spanish Observatorio del Roque de los Muchachos of the Instituto de Astrofísica de Canarias, on the island of La Palma.

VSD, ULTRACAM and HiPERCAM are funded by the Science and Technology Facilities Council (grant ST/Z000033/1). ARM acknowledges support by the Spanish MINECO grant PID2020-117252GB-I00 and by the AGAUR/Generalitat de Catalunya grant SGR-386/2021. RMO is funded by INTA through grant PRE-OBSERVATORIO. MZ acknowledges financial support from FONDECYT grant number 1221059. IP acknowledges support from a Royal Society University Research Fellowship (URF\R1\231496).

DATA AVAILABILITY

Raw and reduced X-shooter data are available through the ESO archive. Raw and reduced HiPERCAM data are available through the GTC public archive. Raw and reduced ULTRACAM data will be shared on reasonable request to the corresponding author.

REFERENCES

Acton J. S., et al., 2021, MNRAS, 505, 2741
Amaro R. C., et al., 2023, ApJ, 948, 129
Bailer-Jones C. A. L., Rybizki J., Fouesneau M., Demleitner M., Andrae R., 2021, AJ, 161, 147
Bear E., Soker N., 2011, MNRAS, 411, 1792
Bédard A., Bergeron P., Brassard P., Fontaine G., 2020, ApJ, 901, 93
Bellm E. C., et al., 2019, PASP, 131, 018002
Belloni D., Zorotovic M., Schreiber M. R., Parsons S. G., Moe M., Garbutt J. A., 2024, A&A, 686, A61
Bensby T., Feltzing S., Lundström I., 2003, A&A, 410, 527
Brown A. J., et al., 2022, MNRAS, 513, 3050
Brown A. J., et al., 2023, MNRAS, 521, 1880
Burrows A., Heng K., Nampaisarn T., 2011, ApJ, 736, 47
Camacho J., Torres S., García-Berro E., Zorotovic M., Schreiber M. R., Rebassa-Mansergas A., Nebot Gómez-Morán A., Gänsicke B. T., 2014, A&A, 566, A86
Casewell S. L., et al., 2015, MNRAS, 447, 3218
Casewell S. L., et al., 2020a, MNRAS, 497, 3571
Casewell S. L., Debes J., Braker I. P., Cushing M. C., Mace G., Marley M. S., Kirkpatrick J. D., 2020b, MNRAS, 499, 5318
Casewell S. L., et al., 2024, MNRAS, 535, 753
Chamandy L., Blackman E. G., Nordhaus J., Wilson E., 2021, MNRAS, 502, L110
Claeys J. S. W., Pols O. R., Izzard R. G., Vink J., Verbunt F. W. M., 2014, A&A, 563, A83
Claret A., Cukanovaite E., Burdge K., Tremblay P. E., Parsons S., Marsh T. R., 2020, A&A, 634, A93
Cojocaru R., Rebassa-Mansergas A., Torres S., García-Berro E., 2017, MNRAS, 470, 1442
Copperwheat C. M., Marsh T. R., Dhillon V. S., Littlefair S. P., Hickman R., Gänsicke B. T., Southworth J., 2010, MNRAS, 402, 1824
De Rosa R. J., et al., 2014, MNRAS, 437, 1216
Dhillon V. S., et al., 2007, MNRAS, 378, 825
Dhillon V. S., et al., 2021, MNRAS, 507, 350
Edgar R., 2004, New Astron. Rev., 48, 843
El-Badry K., Burdge K. B., van Roestel J., Rodriguez A. C., 2023, The Open Journal of Astrophysics, 6, 33
Farihi J., Becklin E. E., Zuckerman B., 2005, ApJS, 161, 394
Foreman-Mackey D., Hogg D. W., Lang D., Goodman J., 2013, PASP, 125, 306
French J. R., et al., 2024, MNRAS, 534, 2244
Freudling W., Romaniello M., Bramich D. M., Ballester P., Forchi V., García-Dabló C. E., Moehler S., Neeser M. J., 2013, A&A, 559, A96
Gaia Collaboration et al., 2023, A&A, 674, A1
Girven J., Gänsicke B. T., Steeghs D., Koester D., 2011, MNRAS, 417, 1210
Graham M. J., et al., 2019, PASP, 131, 078001
Grether D., Lineweaver C. H., 2006, ApJ, 640, 1051
Grieves N., et al., 2017, MNRAS, 467, 4264
Grieves N., et al., 2021, A&A, 652, A127
Hallakoun N., et al., 2023, Nature Astronomy, 7, 1329
Henderson B. A., et al., 2024, MNRAS, 530, 318
Hollands M. A., Littlefair S. P., Parsons S. G., 2024, MNRAS, 527, 9061
Ivanova N., et al., 2013, A&ARv, 21, 59
Johnson J. A., et al., 2011, ApJ, 730, 79
Keller P. M., Breedt E., Hodgkin S., Belokurov V., Wild J., García-Soriano I., Wise J. L., 2022, MNRAS, 509, 4171
Kiman R., Schmidt S. J., Angus R., Cruz K. L., Faherty J. K., Rice E., 2019, AJ, 157, 231
Knigge C., Baraffe I., Patterson J., 2011, ApJS, 194, 28
Koester D., 2010, Mem. Soc. Astron. Italiana, 81, 921
Komacek T. D., Youdin A. N., 2017, ApJ, 844, 94
Kosakowski A., Kilic M., Brown W. R., Bergeron P., Kupfer T., 2022, MNRAS, 516, 720
Kramer M., Schneider F. R. N., Ohlmann S. T., Geier S., Schaffenroth V., Pakmor R., Röpke F. K., 2020, A&A, 642, A97
Lagos F., Schreiber M. R., Zorotovic M., Gänsicke B. T., Ronco M. P., Hamers A. S., 2021, MNRAS, 501, 676
Lew B. W. P., et al., 2022, AJ, 163, 8
Lin Z., et al., 2023, MNRAS, 523, 6162
Littlefair S. P., et al., 2014, MNRAS, 445, 2106
Longstaff E. S., Casewell S. L., Wynn G. A., Maxted P. F. L., Helling C., 2017, MNRAS, 471, 1728
Lothringer J. D., Casewell S. L., 2020, ApJ, 905, 163
Marley M. S., et al., 2021, ApJ, 920, 85
Masci F. J., et al., 2019, PASP, 131, 018003
McAllister M., et al., 2019, MNRAS, 486, 5535
Méndez A., Rivera-Valentín E. G., 2017, ApJ, 837, L1
Muñoz D. J., Petrovich C., 2020, ApJ, 904, L3
O'Connor C. E., Liu B., Lai D., 2021, MNRAS, 501, 507
Paczynski B., 1976, in Eggleton P., Mitton S., Whelan J., eds, IAU Symposium Vol. 73, Structure and Evolution of Close Binary Systems. p. 75
Page E., et al., 2024, AJ, 167, 109
Pala A. F., et al., 2022, MNRAS, 510, 6110
Parsons S. G., Marsh T. R., Copperwheat C. M., Dhillon V. S., Littlefair S. P., Gänsicke B. T., Hickman R., 2010, MNRAS, 402, 2591
Parsons S. G., et al., 2012a, MNRAS, 419, 304
Parsons S. G., et al., 2012b, MNRAS, 420, 3281
Parsons S. G., et al., 2017a, MNRAS, 470, 4473
Parsons S. G., et al., 2017b, MNRAS, 471, 976
Psaridi A., et al., 2022, A&A, 664, A94
Rebassa-Mansergas A., Solano E., Xu S., Rodrigo C., Jiménez-Esteban F. M., Torres S., 2019, MNRAS, 489, 3990